

Correspondence Matrices, Logical Matrices, and Boolean Quantum-Like Structure

Consolidated research findings and corrected status report

Scope

This report consolidates the promising CM/LM findings developed across the September 2026 research threads, with particular attention to the rotation/phase algebra, interference, Bell/GHZ/contextuality, teleportation, dense coding, phase kickback, the LM-to-modal bridge, and the distinction between broad Boolean/modal constructions and results that are genuinely CM-specific or potentially novel. It is a synthesis of prior work, not a fresh literature audit or a re-run of every computation.

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Executive summary

The central outcome is not merely that Boolean CM/LM constructions can imitate familiar quantum language. Superposition, possibilistic interference, Bell-type support patterns, GHZ-like contradictions, teleportation, and related effects already have substantial prior art in modal quantum theory and finite-field models. The more promising direction is to identify exactly what additional structure is contributed by the CM rotation algebra, by its chain-ring valuation, by the restricted operation sets actually induced by CMs, and by the choice between shared-module and literal tensor models.

Most important correction

Universal exact teleportation is not simply impossible. The research passed through several models. Under general invertible Boolean-linear dynamics, exact transfer was first found possible. A later audit showed that this does not mean it is possible under a strict Boolean-orthogonal/unitary-only restriction. In a literal independent-register model, an early four-outcome protocol failed, but exact universal teleportation was restored with a larger measurement/correction scheme and exhaustive verification. The clean resource statement became: universal exact teleportation is possible when the relevant correlated resource map is full rank/invertible and the operation class permits the required linear corrections. Singular resources fail universal transfer. In stricter shared-ring or CM-only formulations, closure and canonical-measurement issues remain.

- **Phase algebra.** The four CM cell positions support a cyclic rotation R with $R^4 = I$. Adding convolution/cyclic multiplication yields the classical algebra $F_2[C_4]$, equivalently $F_2[u]/(u^4)$ with $u = R + I$. This algebra is a finite local chain ring with a nilpotent filtration, and the phase/interference behavior belongs to this added algebraic structure rather than ordinary 2×2 CM multiplication.
- **Interference.** Interference in the Boolean setting is exact XOR cancellation. It is real as an algebraic cancellation phenomenon, but it is not a Born-rule amplitude theory; the probability layer is missing and some natural quadratic/transpose norm attempts are degenerate.
- **Bell/GHZ/contextuality.** Possibilistic Bell, GHZ, Hardy and contextuality phenomena can occur, but broad existence claims are not strong novelty because related finite-field/modal constructions are known. The stronger result is a classification program over the chain ring, especially Smith/valuation criteria and uniform Hardy witnesses for restricted measurement families.
- **Teleportation and dense coding.** Exact protocols are controlled by algebraic rank/invertibility and by tensor/operation choices. Dense coding and teleportation separate: some resources support multiple-message dense coding while still failing universal teleportation.
- **LM bridge.** LM valuation naturally preserves symbolic Boolean contraction and can be enriched to carry A -valued coefficients, but bare LM semantics do not canonically generate

the modal measurement theory. “Nonzero = possible,” modal bases/effects, tensor updates and state-collapse rules are additional contracts.

- **Best research thesis.** Treat the system as a Boolean/modal laboratory with an explicit resource hierarchy: determine which algebraic ingredients are necessary and sufficient for each quantum-information-like capability, and identify counterexamples to false implication chains.

1. Core algebraic setting

1.1 Compact CMs and the rotation lift

A binary logical connective is represented by a compact 2×2 Correspondence Matrix [Theta]. The later quantum-like constructions do not arise merely from multiplying these compact 2×2 matrices. Instead, the four truth-table cell positions are treated as a four-component coefficient space and equipped with a cyclic rotation operator R .

$$R^4 = I$$

Writing the coefficient lift as $\text{Lambda}(a) = a_0 I + a_1 R + a_2 R^2 + a_3 R^3$ over F_2 produces the cyclic group algebra

$$A = F_2[R] \simeq F_2[C_4] \simeq F_2[u]/(u^4), \quad u = R + I.$$

Because characteristic two turns $R^4 - I$ into $(R + I)^4$, the element u is nilpotent of index four. This makes A a finite local chain ring with ideals generated by successive powers of u . The resulting valuation/filtration became one of the most useful organizing devices in the later work.

1.2 What is CM-specific and what is not

An important conceptual correction was to distinguish three layers that had sometimes been blended together:

- **Boolean logic/LM layer.** Truth functions, symbolic formulas, evaluation, contraction and logical pairing.
- **Rotation/chain-ring layer.** The cyclic R action on the four CM positions, its algebra A , units, zero divisors, nilpotents, valuation and Smith normal forms.
- **Modal information layer.** States, effects, tensor composition, possibility semantics, measurement families, corrections and protocol rules.

Only the second layer is directly tied to the chosen CM cell rotation. The third requires additional modelling choices. Consequently, successful Bell or teleportation constructions over F_2 are not automatically evidence of uniquely CM-specific structure.

2. Phase structure and interference

2.1 Phase-like structure

The order-four rotation R provides an intrinsic cyclic phase-like action. Within A , one can classify units, nilpotent elements and valuation classes. The operation resembles a discrete

phase degree of freedom in the sense that powers of R transform components while preserving the cyclic algebraic organization.

The work also identified an analogue of phase reversal through operand/cell reversal, schematically $K R K = R^{-1}$, and related complement behavior to the top nilpotent layer. These are structurally interesting because they tie logical symmetries to the internal algebra generated by cell rotation.

2.2 Interference as XOR cancellation

Once branches are A-valued or represented in an XOR-linear state space, two contributions can cancel exactly because $1 + 1 = 0$ in F_2 . This gives a sharp, computable interference analogue: paths that separately contribute a nonzero component may cancel when recombined.

$$\text{branch}_1 + \text{branch}_2 = 0 \quad (\text{XOR cancellation})$$

This is a genuine algebraic interference phenomenon, but its interpretation must remain modest. It does not by itself produce complex amplitudes, positive probabilities, or the Born rule. Several natural attempts to treat transpose-orthogonality or quadratic forms as a probability analogue were found too degenerate in characteristic two.

Novelty caution

Superposition and cancellation interference over finite fields are not, by themselves, novel quantum foundations claims. The potentially distinctive contribution is the derivation of a particular phase/valuation algebra from literal CM-cell rotation and the way that algebra constrains the allowed operations and resource classes.

3. Superposition, modal semantics, and measurement

XOR-linear combinations provide a natural notion of superposition in the state space. In a modal interpretation, the usual move is possibilistic: an outcome is possible when its amplitude/evaluation is nonzero and impossible when it is zero. Prior art in modal quantum theory means this general idea must be treated as background rather than a headline novelty claim.

3.1 LM-to-modal bridge: what is canonical

- **Canonical or close to canonical.** Boolean valuation of formula-valued LMs; ordinary contraction/evaluation; symbolic support predicates; tensoring or contraction at the symbolic Boolean level when definitions are fixed.
- **Requires added structure.** A-valued amplitudes, “nonzero = possible,” the choice of modal bases/effects, state-update maps, entanglement semantics, and the full measurement family.
- **Consequence.** LMs can encode modal effects after basis choices have been made, but the bare LM calculus does not itself generate the entire modal measurement theory or its previously enumerated basis family.

A useful enrichment was to consider a symbolic Boolean module tensored with A . This preserves exact valuation-wise support information, but it should be described as an enrichment of the LM framework, not as something forced by LM semantics alone.

4. Bell, GHZ, Hardy, and contextuality

4.1 Bell-type behavior

The investigations produced Bell-style possible/impossible support patterns in finite Boolean/modal models. However, two qualifications became central. First, modal Bell phenomena over finite fields are known in prior art. Second, a tested support pattern need not admit a faithful no-signalling probability model; therefore a possibilistic contradiction should not be promoted into a probabilistic quantum claim.

4.2 GHZ

GHZ-like nonseparability and modal contradiction patterns were also explored. A later capability audit found that the details depend strongly on the available measurement family. In particular, there are restricted-setting obstructions: not every familiar complex-Hilbert-space GHZ construction has a direct Boolean counterpart under the same small menu of settings.

4.3 Stronger contextuality direction

The most promising result in this cluster was not “contextuality exists,” but a classification program over the chain ring A . The work identified Smith/valuation data as a resource invariant and developed a residue-hyperplane criterion together with uniform Hardy-style witnesses for restricted measurement families. This is closer to a theorem-level structural contribution than the generic observation that finite-field modal theories can be contextual.

Promising theorem direction

Classify contextuality by the Smith type or valuation profile of the relevant state/resource matrix, while stating the measurement family explicitly. This avoids overclaiming a basis-independent classification and turns the chain-ring algebra into the main mathematical content.

5. Teleportation: corrected chronology

This was the part of the research where the conclusion changed most, because “teleportation” was being tested under different tensor models and different allowed correction groups. The final picture is therefore conditional rather than a single yes/no statement.

Stage	Model / restriction	Finding
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Early broad construction	General invertible Boolean-linear dynamics	Universal exact transfer appeared possible.
Restriction audit	Boolean-orthogonal / unitary-like operations only	The broad protocol does not imply teleportation here; the needed corrections need not lie in this smaller group.
Literal independent-register audit	Literal F2 tensor/register model	An old four-outcome construction failed, but a larger exact protocol restored universal transfer and was exhaustively checked.
Resource criterion	Literal/general linear setting	Universal exact teleportation iff the correlated resource map is full rank/invertible (rank 8 in the audited encoding). Singular resources fail universal transfer.
Strict shared-ring / CM-only view	A-module/shared-module operations with restricted intrinsic gates	Closure, tensor-choice and canonical-measurement issues remain; universal teleportation should not be advertised here without an explicit operation set and proof.

5.1 The clean algebraic statement

In the broad literal Boolean-linear model, teleportation reduces to a rank/invertibility condition on the resource branch together with the availability of branch-dependent corrections. If the branch map from the unknown input to the receiver is invertible, the receiver can undo it. If the resource is singular, no correction can recover every possible input because information has already been lost.

universal exact transfer $\Leftrightarrow$ every accepted branch map is invertible

In the audited encoding this was summarized as a full-rank resource criterion (rank 8 for the relevant matrix representation). One implementation used a 64-outcome construction and was checked exhaustively over 16,384 cases.

5.2 Why the “impossible” memory is still understandable

Several narrower statements really are negative. A protocol may fail when corrections are restricted to an orthogonal/unitary-like subgroup; a shared-module construction may not represent independent tensor factors faithfully; a singular resource cannot support universal transfer; and an LM/CM-only theory does not automatically supply the measurement/update

structure needed for teleportation. Thus “teleportation is impossible” was a reasonable interim summary for one restricted model, but it is not the final cross-model conclusion.

5.3 Multi-copy activation

A further result was that a resource that is insufficient in one copy can become useful when multiple copies and broader Boolean-linear filtering are permitted. The example discussed in the audit included a singular Smith type that fails singly yet admits heralded activation in a two-copy enlarged literal model. This is potentially interesting as a finite-ring resource phenomenon, but it must be stated with its generous operation class.

6. Dense coding and separation of capabilities

Dense coding proved useful because it does not track teleportation one-for-one. Some shared-module resources supported several distinguishable messages even though they were not universal teleportation resources. This gives explicit separations between operational capabilities and supports the broader “resource hierarchy” viewpoint.

- **Dense coding can survive without universal teleportation.** At least one noninvertible/valuation-class resource supported a four-message dense-coding behavior while still failing the universal-transfer criterion.
- **Invertibility matters differently across tasks.** Teleportation asks for recovery of an arbitrary unknown input, so branch invertibility is decisive. Dense coding only asks for distinguishability of an encoded finite message set, which is weaker.
- **Research implication.** The correct object may be a capability lattice rather than a single scalar notion of “quantumness.”

7. Phase kickback and Fourier obstruction

The capability audit found a useful asymmetry. A form of shared-module phase kickback can be constructed using the A-valued phase action, yet the natural Boolean analogue of a Fourier/Hadamard transform can be singular. One tested transform had Smith form $\text{diag}(1, u^2)$, so it is not invertible.

$$\text{Smith}(H_z) = \text{diag}(1, u^2) \Rightarrow H_z \text{ is singular}$$

This matters because it shows that “phase behavior” does not force the rest of the usual complex quantum toolkit. It supplies another clean separation: phase kickback-like behavior can exist even when there is no invertible Boolean Fourier transform of the desired kind.

8. Tensor choice: shared module versus literal tensor product

One of the most important corrections was the recognition that two constructions had sometimes been conflated:

- **Shared-module model.** Multiple logical/phase degrees of freedom are represented inside a common A-module or related structured space. This is compact and often preserves the CM-derived algebra, but it does not automatically behave like independent subsystems.

- **Literal tensor/register model.** Independent registers are represented by an actual tensor/product construction over F_2 (or the chosen coefficient structure). This is the natural setting for statements that require subsystem independence, but it can enlarge the operation set beyond what is intrinsically generated by CMs.

Several teleportation and entanglement claims changed when this distinction was enforced. Any future paper should declare the tensor model at the beginning of each theorem or protocol.

9. What appears genuinely promising

Topic	Status	Why it matters
CM rotation algebra $A = F_2[C_4] = F_2[u]/(u^4)$	Solid algebraic foundation	Provides a concrete chain-ring phase/valuation structure derived from CM cell rotation.
Nilpotent filtration and valuation resources	Promising	Gives a hierarchy finer than rank alone and connects directly to Smith normal form.
Smith/valuation contextuality classification	Strongest theorem-like direction	Moves beyond existence of modal contextuality toward structural classification under explicit measurement families.
Capability separations	Promising	Dense coding without universal teleportation; phase kickback without invertible Fourier transform; singular-vs-invertible resource distinctions.
Teleportation rank criterion	Useful, but not uniquely CM	Clean theorem in the broad Boolean-linear model; value comes from integration with CM-derived resource classes and restrictions.
LM symbolic-to-modal enrichment	Conceptually useful	Clarifies exactly what LMs supply and what must be added; prevents category mistakes and overclaiming.
Generic superposition/Bell/GHZ/teleportation existence	Background / validation	Interesting demonstrations, but heavily anticipated by modal quantum and finite-field prior art.

10. Results that should be presented cautiously

- **“Quantum superposition from CMs.”** Better phrased as XOR-linear/modal superposition inside a CM-derived or CM-enriched algebraic model; finite-field/modal superposition itself is established prior art.
- **“CMs prove Bell nonlocality.”** Too strong. What was demonstrated is possibilistic Bell-style incompatibility/support behavior in specified models and measurement families, without a Born-rule probability layer.
- **“CMs have unitary quantum dynamics.”** Not in the ordinary complex-Hilbert sense. Characteristic-two orthogonality is restrictive/degenerate, and many useful Boolean-linear corrections are not orthogonal.
- **“Universal teleportation is impossible.”** Too broad. It fails in some restricted models, but is possible in a larger literal Boolean-linear setting with full-rank resources and appropriate corrections.
- **“LM semantics generates the modal measurement theory.”** Not canonically. Measurement bases, possibility semantics and update rules are additional structures.
- **“One tensor construction fits all claims.”** False. Shared-module and literal-register/tensor models answer different questions and must be separated.

11. Recommended research program

The strongest continuation is to convert the collection of examples into a formal capability/dependency theory. Instead of asking whether the formalism can “do quantum mechanics,” ask which algebraic resources are necessary and sufficient for each phenomenon.

phase structure $\Rightarrow$ interference $\Rightarrow$ nonseparability $\Rightarrow$ contextuality

invertible correlated resource $\Leftrightarrow$ universal exact teleportation (within the broad linear model)

- **Build an implication/counterexample lattice.** For every pair of capabilities, prove implication, disprove it by an explicit resource, or mark it open.
- **Parameterize by valuation/Smith type.** Use units, zero divisors, nilpotent depth, rank and Smith normal form as resource descriptors.
- **Separate operation classes.** At minimum distinguish unrestricted linear, invertible linear, Boolean-orthogonal/unitary-like, CM-generated, and measurement/filter-augmented operations.
- **Separate subsystem models.** Repeat key theorems in both shared-module and literal-tensor settings when meaningful.
- **State measurement assumptions locally.** Contextuality and teleportation theorems should name the basis/effect family rather than relying on an implicit universal measurement theory.
- **Audit novelty theorem-by-theorem.** Use modal quantum theory, finite-field quantum models, finite-chain-ring linear algebra, contextuality, process theories and algebraic coding/resource literature as comparison classes.

12. Suggested paper architecture

A coherent paper could be organized around resources and separations rather than around a catalogue of quantum analogues:

- Foundations: compact CMs/LMs, the four-cell rotation, $A = F_2[u]/(u^4)$, valuation and Smith form.
- Model declarations: state spaces, shared-module versus literal tensor composition, admissible operations and measurement semantics.
- Phase/interference section: exact algebraic cancellation, phase reversal, Fourier obstruction.
- Resource classification: units, zero divisors, nilpotent layers, rank and Smith type.
- Protocols: dense coding and exact teleportation with explicit operation assumptions.
- Contextuality: Bell/GHZ/Hardy demonstrations followed by the stronger Smith/valuation classification result.
- Separation theorems: dense coding without teleportation, phase kickback without invertible Fourier transform, contextuality dependence on measurement family.
- Limitations/prior art: no Born rule, no automatic unitary group, no canonical LM measurement theory, and finite-field/modal antecedents.

13. Provenance and chronology of the consolidated findings

This report consolidates conclusions from the September 2026 CM/LM research sequence. The most relevant stages were:

Date	Research stage	Consolidated contribution
17 Sep 2026	Phase-calculus / pure-Boolean investigation	Established the CM rotation algebra, interference constructions and initial quantum-information analogues; later audits narrowed novelty claims.
17 Sep 2026	Teleportation and correctness audits	Corrected the early protocol, separated shared-module from literal registers, and obtained the full-rank universal-transfer criterion in the broader model.
17-18 Sep 2026	Contextuality and chain-ring investigations	Elevated Smith/valuation classification and uniform Hardy-family ideas above generic “contextuality exists” claims.
18 Sep 2026	LM-to-modal bridge study	Showed that symbolic LM valuation is natural but the modal measurement/effect layer is an added contract, not a canonical consequence.

18 Sep 2026	Capability/dependency theory	Organized dense coding, teleportation, phase kickback, Fourier obstruction and valuation resources into a resource-separation program.
21-22 Sep 2026	Foundations/computational paper split	Separated the Boolean operator-calculus foundations from computational/compiler work; quantum/modal material remained a distinct research line rather than being folded casually into the foundations manuscript.

14. Bottom line

The most defensible and interesting story is not that CMs reproduce quantum mechanics. It is that the CM four-cell rotation leads to a concrete finite chain-ring algebra whose valuation, nilpotent structure and restricted operation classes create a small laboratory in which familiar information-theoretic phenomena can be separated, classified and compared. The best next results are likely to be structural: necessary/sufficient conditions, capability separations, and classification theorems tied to valuation/Smith data.

Current teleportation status in one sentence

Universal exact teleportation is achievable in the broader literal Boolean-linear model with a full-rank resource and sufficiently general branch corrections, but it is not thereby established for a strict CM-generated/shared-module or orthogonal-only theory.

Appendix A. Compact status matrix

Claim / capability	Status	Qualification
XOR superposition	Yes	Known modal/finite-field phenomenon; not headline novelty.
CM-derived phase algebra	Yes	$A = F2[C4] = F2[u]/(u^4)$; important structural core.
XOR interference	Yes	Exact cancellation; no Born rule.

Bell-style possibilistic contradiction	Yes in specified models	Prior art exists; probability/no-signalling caveats.
GHZ-like behavior	Yes in some settings	Measurement-family restrictions and obstructions matter.
Contextuality classification	Promising/strong	Smith/valuation criterion under explicit measurement family.
Dense coding	Yes in tested models	Can occur without universal teleportation.
Universal exact teleportation	Conditional yes	Full-rank/invertible resource + sufficiently general corrections in literal linear model.
Universal teleportation with singular resource	No	Rank loss destroys recoverability; multi-copy activation may alter resource status under broader filters.
Universal teleportation under orthogonal-only corrections	Not established; generally obstructed	Broad linear protocol does not imply this stricter result.
Phase kickback	Yes in shared-module construction	Does not require an invertible Fourier transform.
Invertible Boolean Fourier/Hadamard analogue	Obstructed in tested construction	Natural H_z was singular with Smith $\text{diag}(1, u^2)$.
Canonical LM $\rightarrow$ full modal measurement theory	No	Requires extra basis/effect/possibility/update choices.
Born-rule probability theory	No	Not supplied by the Boolean/modal framework developed so far.